

NETWORK DESIGN AND MANAGEMEN

CONTINUOUS ASSESSMENT TEST (CAT) 1 TERM 3

MARKING SCHEME

Total Marks: 50

QUESTION 1 (4 Marks)

a) Half-duplex (1 Mark)

Communication mode where data can be transmitted in both directions but **not at the same time**.

b) Node (1 Mark)

Any device connected to a network that can send, receive, or process data.

c) Bandwidth (1 Mark)

The maximum amount of data that can be transmitted over a network connection in a given period of time.

d) Topology (1 Mark)

The physical or logical arrangement of devices and communication links in a network.

QUESTION 2 (4 Marks)

Differentiate between LAN and WAN

LAN	WAN
Covers a small geographical area. (1 Mark)	Covers a large geographical area. (1 Mark)
Usually owned by a single organization. (1 Mark)	Often uses public or leased communication links. (1 Mark)

Total: 4 Marks

QUESTION 3 (4 Marks)

Advantages of Star Topology

Award **1 Mark each** (Any Four)

1. Easy to install and manage.
2. Easy to add or remove devices.
3. Failure of one node does not affect others.
4. Easy fault detection and troubleshooting.
5. Better network performance.
6. Centralized management.
7. High reliability.

Total: 4 Marks

QUESTION 4 (10 Marks)

a) Star Topology (2 Marks)

A network topology where all devices are connected to a central device such as a switch or hub.

b) Ring Topology (2 Marks)

A topology where each device is connected to two neighboring devices, forming a circular path for data transmission.

c) Partial Mesh Topology (2 Marks)

A topology where some devices have multiple direct connections while others are connected through intermediary devices.

d) Hybrid Topology (2 Marks)

A combination of two or more network topologies within the same network.

e) Point-to-Point Topology (2 Marks)

A topology where two devices are directly connected through a dedicated communication link.

Total: 10 Marks

QUESTION 5 (4 Marks)

Components of a Computer Network

Award **1 Mark each** (Any Four)

1. Computers/workstations
2. Network Interface Cards (NICs)
3. Switches
4. Routers
5. Transmission media (cables/fiber)
6. Wireless access points
7. Servers
8. Modems

Total: 4 Marks

QUESTION 6 (4 Marks)

Functions of a Router

Award **1 Mark each** (Any Four)

1. Connects different networks.
2. Routes packets between networks.
3. Determines the best path for data transmission.
4. Acts as a default gateway.
5. Performs Network Address Translation (NAT).
6. Provides basic network security through filtering.
7. Connects LANs to the Internet.

Total: 4 Marks

QUESTION 7 (4 Marks)

Differentiate between PAN and MAN

PAN	MAN
Covers a very small area around an individual. (1 Mark)	Covers a city or metropolitan area. (1 Mark)
Examples: Bluetooth, mobile hotspot. (1 Mark)	Examples: City-wide fiber network. (1 Mark)

Total: 4 Marks

QUESTION 8 (6 Marks)

a) Suitable Network Type (3 Marks)

Answer: Local Area Network (LAN) (1 Mark)

Reason: Suitable for connecting computers within a limited area such as a computer laboratory, offering high speed and low cost communication. (2 Marks)

b) Networking Devices Required (3 Marks)

Award 1 Mark each (Any Three)

1. Switch
2. Router
3. Network cables
4. Wireless Access Point
5. Modem
6. Network Interface Card (NIC)

Total: 6 Marks

QUESTION 9 (10 Marks)

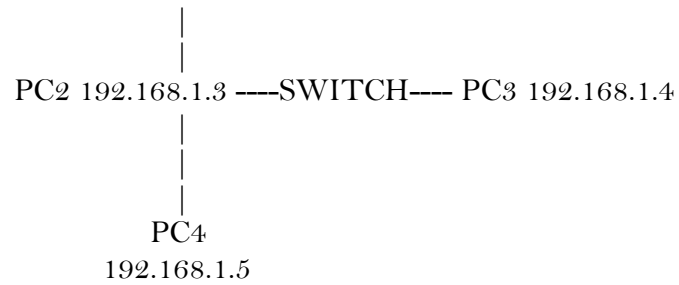
Star Topology Diagram

Mark Allocation

Item	Marks
Correct star topology structure	3
Central switch correctly labeled	2
Four computers correctly connected	2
Correct IP addresses assigned	2
Network information/subnet indicated	1
Total	10

Sample Answer

PC1
192.168.1.2
|



Network Address: 192.168.1.0/24

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

Alternative Marking Guide

- Central Switch = 2 Marks
- Four correctly connected nodes = 2 Marks
- Star arrangement = 3 Marks
- IP addresses = 2 Marks
- Gateway/Subnet information = 1 Mark

Total: 10 Marks